

Phys 247 : Fall 2026

Problem Set 1

Assigned 9/2, Due Monday 9/14

Submit HW by Start of Class, No Late Grades

Start early!

Problem 1 (easy)

Demonstrate that you understand the definitions of vector algebra.

KK 1.1, 1.2, 1.3

Problem 2 (medium)

Apply vector algebra creatively.

KK 1.9

KK 1.11

Problem 3 (medium)

KK 1.17

Hint: “rolling withing slipping” means $\Delta s = R \Delta\theta$, where angle of rotation θ is related to arc length s .

Problem 4 (hard)

KK 1.23 - This is an opportunity to use “jerk”! for maximizing comfort no less.

Hint: for (c) review the sine approximation in Note 1.3 of the textbook.

Extra Credit (hard)

KK 1.26 - I like this problem a lot!

Hint: how might one simplify the problem? what if $\phi=0$?

Hint: Can you find any dimensionless quantities? Does the quantity $\bar{x} = x g / 2v_0^2$ look interesting?

Reminder: to optimize $f(x)$, try setting $df/dx=0$.

Submission Instructions:

Scan a PDF. Either use an office machine or make clear photos with your cell phone.

Upload to canvas. Do **not** use AI to enhance, check, or auto-latex your solutions. You're encouraged to work together in groups, but everyone must write and submit solutions individually.